PathwayAI | Dutchess County

# Lyme Disease Burden & Community Action

Prepared 2026-11-01 · Public-data planning brief · Version 1.3

| Reported cases · 2024 | County rate per 100,000 · 2024 | State rate per 100,000 · 2024 |
|---|---|---|
| 1,060 | 355.5 | 109.8 |

## Illustrative economic burden estimate

# USD 2.89 million

2025 purchasing-power comparison · 11 available CPI months · general inflation

Study-based illustration, not measured county spending or predicted savings. Sources [1, 5, 11]; calculation and methods: page 3.

## Neighboring county and state comparison

| Area | Reported cases · 2024 | Rate per 100,000 · 2024 |
|---|---|---|
| Dutchess | 1,060 | 355.5 |
| Orange | 964 | 237 |
| New York State | 21,632 | 109.8 |

## Healthcare resources and population context

| Indicator | Value | Period |
|---|---|---|
| Primary-care physicians | 233 | 2023 |
| Hospitals | 2 | 2023 |
| Hospital beds | 404 | 2023 |
| Uninsured residents under age 65 | 4.7% | 2022 |
| People below poverty level | 8% | 2023 |
| Language other than English at home · age 5+ | 16.2% | ACS 2020–2024 |

General resource/population context, not Lyme-attributable need, appointment availability or interpreter demand. Selected insurance/poverty fields supply no margin of error.

## Reported Lyme rate over time · per 100,000 residents

| 2022 | 2023 | 2024 |
|---|---|---|
| 394.6 | 358.9 | 355.5 |

County of residence, not exposure location. Reporting years and data sources differ; trends do not establish causes. Detailed references and limitations: page 3.

# From evidence to community action

## Support priority: Getting to care

Question: Should a county care-navigation team investigate a small ride-navigation pilot with willing residents?

Possible response: Ask willing residents and transport providers to review ride eligibility, accessibility and appointment coordination.

Who could help: Patient and caregiver representatives, transport services and care navigators

## Actions to discuss · Public health departments

- Share tick-prevention information through local events and trusted community channels.
- Review county surveillance alongside service availability; ask residents which barriers need attention.
- Bring residents and potential partners together to choose one support priority and agree how to follow up.

## Four areas for community action

Prevention education and tick-bite awareness: Share established prevention information through trusted local channels.

Timely access to healthcare: Help residents find care and ask about appointments, costs, transport and interpreters.

Financial and caregiving burden: Recognize spending, missed work, unpaid care and support needs beyond medical bills.

Language access and community resources: Make information easier to understand and connect residents with existing local support.

## Resident priorities to discuss

Confirm priorities with residents and caregivers before choosing a response. No county-specific patient-reported findings are established in this brief.

## Before starting a pilot

- Confirm local demand with residents and caregivers, including assisted/offline participation.
- Verify current services, eligibility, interpreter access and appointment capacity.
- Agree who is responsible and estimate staff time, program costs and a funding source.
- Define consent, minimal data collection, access, retention and deletion before any collection.

## Check whether people received support

Choose a common follow-up period, such as 30 days. Record the starting situation, help received and needs still unmet. Keep unanswered questions unknown.

For this priority: Requested and completed accessible rides; appointments attended; travel spending; support still needed

Further evaluation is needed to establish health benefits or savings. Actions are proposals for local review, not adopted programs or commitments.

# References & methods

## Original study basis

1,060 reported cases × USD 2,032 study mean = USD 2,153,920 (2016 dollars).

Hook et al. (published 2022): 613 participants in societal analysis; reported cases recruited in four high-incidence states during 2014–2016. Societal: medical, nonmedical and productivity costs; components are already included.

Applying this study mean to reported county cases is an assumption; the result is not measured local spending. No adjustment for underreporting or local case mix. No causal savings or disability probabilities are inferred.

Hook excluded non-English speakers and nonacute disease. The study does not represent every local household or continuing lifetime costs.

## Optional inflation comparison

General CPI factor approximately 1.34; calculations use the full-precision factor retained in the methods CSV. Adjusted study mean approximately USD 2,726 per participant (rounded for display).

2016 uses 12 monthly observations; 2025 uses 11 available months. October is missing; no imputation. This is not a verified official full-year annual average or medical-care inflation adjustment. It changes purchasing power, not case mix, treatment patterns or local prices.

## Source register


1. Reported cases, county of residence · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/cases.pdf

2. Reported rates per 100,000 · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/rates.pdf

3. CDC surveillance interpretation; historical county file in app · 2019–2022 county file; separate from NYSDOH 2024

https://www.cdc.gov/lyme/data-research/facts-stats/index.html

4. Healthcare resources, insurance and poverty context · 2022 insurance fields; 2023 healthcare and poverty fields; AHRF 2024–2025 release

https://data.hrsa.gov/topics/health-workforce/nchwa/ahrf

5. Societal study mean · 2016 dollars; recruitment 2014–2016

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

6. Reported rate trend, Figure 147, page 219 · 2022–2023; regional assessment published 2025

https://www.dutchessny.gov/Departments/DBCH/Docs/MHRCHA2025.pdf

7. Language spoken at home, age 5+ · ACS 2020–2024

https://www.census.gov/quickfacts/dutchesscountynewyork

8. Tick Task Force · Existing directory; current status requires verification

https://www.dutchessny.gov/Departments/County-Legislature/Tick-Task-Force.htm

9. Mobile Health Unit · Existing directory; current status requires verification

https://www.dutchessny.gov/Departments/DBCH/Mobile-Health-Unit.htm

10. Community Health Improvement Plan · Existing directory; current status requires verification

https://www.dutchessny.gov/Departments/DBCH/Community-Health-Improvement-Plan.htm

11. BLS CPI-U via supplied FRED file · 2016 and 2025 available-month comparison

https://fred.stlouisfed.org/series/CPIAUCNS


## Verification and interpretation

Existing source snapshots; newer official releases have not been verified for this brief. Official rates are reproduced; exact rate-denominator vintage is not independently verified. Pre-2022 rates are excluded because surveillance definitions changed.

Census language context is an existing QuickFacts snapshot, not a fresh pull or a measure of English proficiency. HRSA field years are not substituted with newer Census periods. Initiative links require current local verification; no listing does not mean no services exist.

Public evidence and proposed planning actions only. No session stories, personal records or synthetic patient cards are included.

Discussion brief, not a county demand assessment, benefits decision, funded service or adopted county plan. No county or organizational endorsement is implied.